

Commitment and Participation: The Minimal Ontology of Event Density

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Series: Event Density (ED) Generative Papers — substrate-evaluation arc (primitive minimality / reduction) **Status:** Publication draft (structural). A minimality audit of the canonical 13 primitives; consolidates the primitive-reduction program. Standalone; cold-reader accessible. **Repository target:** physics-papers/substrate-evaluation/ (ED-Generative) — substrate-evaluation

Preamble: What This Paper Does NOT Claim (*written first per QC discipline; abstract reconciled against this*)

1. **The primitives are not derived from nothing.** This paper does not derive ED’s substrate from a deeper layer; it audits the *internal* structure of the canonical 13 (Paper_087) and finds that some are facets of others. The postulational genre is unchanged.
 2. **This is NOT a claim that ED reduces to one primitive.** The honest result is a *pair* — commitment and participation — over a spatial arena. Commitment cannot swallow the things it acts on (participation, the channels, the amplitude) or the spatial stage. Any reading of this paper as “ED is just the arrow” is a misreading; the paper argues the opposite where the arrow overreaches.
 3. **Three-dimensionality is not derived here.** P06 stays a primitive/selection. The paper reports a *conditional* derivation candidate (via linking) and is explicit that its premise is open.
 4. **The “arrow as keystone” synthesis is an account, not a theorem.** That commitment does many of physics’ jobs is a unifying observation with per-job support elsewhere in the program; it is not a proof that the other primitives are theorems of commitment.
 5. **Empirical values remain inherited.** The reduction is of the primitive *structure* (which primitives are independent), not of the value layer (masses, couplings, cosmological parameters), which is inherited from measurement throughout ED.
 6. **No new primitive or rule is introduced.** Every move is an analysis of the existing canonical 13; nothing is added to the ontology.
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Abstract

Event Density postulates thirteen substrate primitives (Paper_087). This paper asks the minimality question the position paper itself flags (§8.4): does the set contain a derivable subset? We run a structured reduction audit — for each primitive, is it a **facet** of another primitive, **presupposed** by it (logically prior), or **independent**? — and reach a compact result. (1) The primitives split

cleanly into three roles: the *things* commitment acts on (participation, distinguishable channels, a complex amplitude), the *arena* it runs on (spatial homogeneity, three dimensions, scale, kinds), and *commitment itself* with its facets (forward-causal chains, polarity-transport, the stability landscape’s descent-*direction*, and — reclassified here — time-homogeneity as a process-side property). Commitment (P11) is the sole **process**: every dynamical and directional structure in ED is commitment acting. **(2)** One reduction and one reclassification tighten the count. The bandwidth and phase primitives **fuse** into one complex amplitude — grounded on the result (inherited from the quantum-logic reconstruction) that the amplitude field is $\mathbb{C}$, so bandwidth and phase are two real readings of one complex object rather than two primitives. And **time-homogeneity is reclassified from the arena to the process**: canonical P13 is stated as a predicate *of* the dynamical primitive (the process-law is time-translation invariant), so it is a property of commitment’s one law, not an independent stage-fact. This is a reclassification, not a derivation of P13 from commitment’s collapse-content — the position paper factors P13 out precisely because P11 does not entail it. **(3)** Time-homogeneity does *not*, however, merge with spatial homogeneity into one spacetime symmetry: ED’s time carries the arrow and its space does not, so no unified translation symmetry exists at the substrate level — Poincaré invariance is emergent. The net honest floor is **3 substrate + 4 arena + 1 commitment** (with P12’s descent-direction, but not its functional form, on the commitment side). Its shortest *naming* is the pair **commitment and participation** — the sole process and the primitive relation it acts on — offered as a mnemonic, not as a claim that the eight-item floor collapses to two. The capstone: ED has a spatial stage but **no separate temporal stage** — space is a genuine arena, while time’s aspects (direction and law-invariance) are both properties of the process. Three-dimensionality is the one arena member with a live (conditional) route back to commitment, via a linking argument whose premise remains open. Throughout, the reduction is of primitive *structure*, not empirical values, and it introduces no new primitive.

1. Introduction

Event Density is a substrate ontology built on thirteen postulated primitives, enumerated authoritatively in the position paper (Paper_087). The position paper is careful to state that the set is postulated, not derived, and it flags its own minimality as open — §8.4 offers a falsifier sentence: “*Demonstration that the canonical 13 contains a derivable subset (e.g., P13 derivable from P11 + P03, or P06 derivable from P03 + P07) would reduce the primitive count without changing downstream content.*”

This paper takes up that invitation and reports the result of a systematic reduction audit. The question is not whether the primitives can be derived from *outside* ED (they are the foundational layer, by construction) but whether, *inside* the set, some primitives are facets of others. The answer is yes — at least one genuine reduction exists (two substrate primitives fuse) — and the shape of the answer is more informative than the count: the primitives organize into three roles, one of which, commitment, is (under a pivot the paper discloses) the only source of process in the theory. It accounts for the entire temporal side — time’s aspects turn out to be properties of the process, not arena facts — while leaving the spatial side, and its own nouns, genuinely irreducible.

The result gives ED a compact, honest description. It is not “ED is one primitive” (that is false, and the paper says where it fails). Its shortest *naming* is a pair — commitment, and the participation it acts on, over a spatial stage — offered as a mnemonic for a fuller eight-item floor. Section 2 states the method and its pivot caveat. Sections 3–6 run the audit, the one reduction, and the

one reclassification. Section 7 states the floor and the capstone. Sections 8–10 give the arrow-as-keystone synthesis, limitations, and falsifiers.

2. Method: The Reduction Audit

The primitives (Paper_087 §5, abbreviated): P01 event-density-layer existence; P02 participation as primitive relation; P03 channel + locus indexing with spatial homogeneity; P04 bandwidth (non-negative additive scalar); P05 polarity-transport along edges; P06 spatial dimension ($D = 3+1$); P07 distinguishable channels; P08 substrate scale ℓ_{ED} ; P09 polarity (U(1) phase); P10 rule-type multiplicity; P11 commitment irreversibility; P12 stability landscape; P13 time homogeneity.

For each primitive we ask, against the canonical statement of commitment (P11: “*at commitment events, a chain’s multi-channel participation collapses to single-channel, with the un-selected channels’ phase content randomized; irreversibly*”):

- **FACET** — definitionally part of, or entailed by, another primitive $\implies$ it reduces.
- **PRESUPPOSED** — the other primitive’s definition already requires it $\implies$ it is logically *prior* and cannot reduce to that primitive (the primitive needs it to be stated at all).
- **INDEPENDENT** — neither entailed by nor presupposed by the others.

The test is deliberately conservative: “does many jobs” is not “is the only primitive,” and a primitive that P11’s own definition references cannot be a facet of P11 — it is prior to it. The audit is a structural analysis of the primitive statements, not an empirical experiment; where a step rests on a measured or separately-derived result, that is flagged in the load-bearing table (§2.5).

A methodological caveat, stated up front. This audit is *P11-pivoted by construction*: it tests each primitive’s reducibility *relative to* commitment, not all pairwise dependencies among the thirteen. Such a test cannot, by itself, establish that commitment rather than some other primitive is the natural pivot — that choice is an *input*, motivated by the keystone pattern (§8), not an output of the audit. What the audit *can* do is show where commitment succeeds and, more importantly, where it fails to absorb a primitive (its own nouns, its spatial stage); those failures are genuine findings independent of the pivot choice. Read the “sole process” result (§3) accordingly: it is what the P11-pivoted audit returns, with the pivot disclosed, not a neutral proof that no other organizing primitive is possible.

2.5 Load-Bearing Step Audit

Step	Status	Source / justification
13 primitives postulated	P	Paper_087
Audit is P11-pivoted (tests reducibility <i>relative to</i> commitment)	method caveat	§2 — an input motivated by the keystone pattern, not a neutral all-pairs proof
Three-role split (nouns / arena / commitment+facets)	structural	§3 — audit against P11’s definition
Commitment (P11) is the sole <i>process</i> primitive	structural	§3 — dynamical/directional content is P11 acting; P01-ordering identification flagged interpretive

Step	Status	Source / justification
P01/P05 are commitment-activated facets	structural	§3 — forward-causality, transport-direction
P12: descent- <i>direction</i> is a facet; functional <i>form</i> (Σ signs) is independent	structural	§3 — the arrow supplies the motion, not the law
Bandwidth + phase fuse into one complex amplitude	D (grounded, inherited)	§4 — grounded on the C-field result (from the reconstruction paper, as strong as it)
Channels (P07) and participation (P02) stay distinct	structural	§4 — basis and relation are not the amplitude value
Time-homogeneity (P13) is a property of the process, not an arena primitive	structural (reclassification)	§5 — canonical P13 is predicated <i>of</i> the dynamical primitive; NOT derived from P11’s collapse-content
P03 and P13 do NOT merge into one spacetime homogeneity	structural	§5 — time is arrowed, space is not (rests on P11 being the sole asymmetric axis)
Three-dimensionality (P06) is derived	NOT CLAIMED	§6 — selected; two of three landings read 3 off observed content, one is a conditional candidate
“Order is not label-held” (3D premise input)	measured	§6 — Kendall $\tau \approx 0.96$ vs 0.5, certified sim (rules out labels; does not by itself establish linking)
Arrow does many of physics’ jobs (keystone synthesis)	interpretation	§8 — per-job support cited, not a theorem; does not compound
Empirical values (masses, couplings) reduced	NOT CLAIMED	preamble 5 — value layer inherited throughout

Every step is P, method-caveat, structural, D, measured, interpretation, or explicitly not-claimed. Note the honest tiering of the two count-reducing moves: the amplitude fusion is **D** but *inherited* (only as strong as the C-field result it rests on); the time-homogeneity move is a **reclassification** (P13 belongs with the process, not the arena), **not** a derivation of P13 from commitment. Three-dimensionality is explicitly **not** claimed derived.

3. The Three-Role Split, and Commitment as the Sole Process

The audit produces a clean three-way partition.

Role 1 — the things commitment acts on (PRESUPPOSED by P11, hence prior): P02 participation (P11 collapses *participation*), P04 bandwidth (the collapse is bandwidth-weighted — the Born-rule content), P07 channels (P11 collapses *to a channel*, among *distinguishable* ones), P09

phase (P11 *randomizes un-selected phases*). These four are exactly the objects P11's own definition references; commitment cannot reduce them, it presupposes them. They are also, together, the participation amplitude $P_K = \sqrt{b_K} e^{-i\pi_K}$ on channel K — commitment is what commits it.

Role 2 — the arena (INDEPENDENT of the arrow): P03 spatial homogeneity, P06 three dimensions, P08 scale, P10 rule-type multiplicity, and (before §5) P13 time-homogeneity. These are the stage the process runs on — spatial, scale, and kind facts the arrow neither creates nor presupposes.

Role 3 — commitment and its facets (the process): P11 commitment itself; P01 chains, whose forward-directedness is the arrow applied to participation-events; P05 polarity-transport, whose forward-directedness is the arrow carrying the phase; and P12 the stability landscape, but only *in part* — its descent-*direction* (that the chain moves forward down Σ) is commitment's, while its functional *form* ($\Sigma = \text{Coh} - \text{Str} - \text{Grad}$, and the descent signs) is an independent specification that commitment does not supply. So P12 is a partial facet: the arrow provides the motion, not the law.

One interpretive seam, flagged. Ruling P01 (chains) a facet identifies its *forward-causal ordering* with P11's *collapse-irreversibility*. Causal precedence (a partial order on events) and irreversibility (no map from post-state to pre-state) are, in general physics, separable notions; ED treats them as the one arrow, but that identification is an interpretive merge, not a forced entailment. It is tiered structural-with-an-open-question, not settled — a reader who holds causal order distinct from irreversibility would count P01's ordering as its own primitive.

The refined statement, which the rest of the paper sharpens:

Commitment (P11) is the only PROCESS primitive. Everything static — the nouns it acts on and the arena it runs on — is not commitment and does not reduce to it. But everything dynamical or directional — forward-causal chains, polarity-transport, the descent of the stability landscape, and (as physics) measurement and the pointer basis, matter/antimatter, gravity's causal direction, information preservation, the hypercharge reference — *is* commitment acting. Commitment is the single source of process; the other primitives are its objects and its stage.

This is why commitment appears everywhere in ED: not because it is the only primitive, but because it is the only *process*, so any dynamical structure, followed to its root, is commitment.

4. First Reduction: Bandwidth and Phase Fuse into One Complex Amplitude

The four Role-1 nouns look like four primitives, but two of them fuse. Bandwidth $b_K = |P_K|^2$ and phase $\pi_K = \arg P_K$ are the modulus and phase of one complex number $P_K \in \mathbb{C}$. In isolation that is only polar coordinates, not a reduction (the complex number still has two real degrees of freedom — the fusion reduces the *primitive count*, not the degrees of freedom).

What carries the reduction is one result, inherited: the amplitude field is $\mathbb{C}$. ED's amplitude field is complex — the U(1) phase (P09) rules out a real field, and composite structure rules out quaternionic (this is the $\mathbb{C}$ -selection of the quantum-logic reconstruction paper, and the fusion is only as strong as that result, which is itself reconstruction-tier). *Because* the fundamental object is a single $\mathbb{C}$ -valued amplitude, bandwidth and phase are two real *readings* of that one object,

not two independently-postulated primitives. That is the $2 \rightarrow 1$: one complex primitive replaces a separate real-modulus primitive and real-phase primitive.

Complementarity — that a which-channel (bandwidth) commitment destroys phase coherence, and a definite relative phase forces equal bandwidths — *corroborates* this (the two readings cannot be independently dialed), but it does not by itself do the reducing: position and momentum are complementary yet nobody counts space as one primitive rather than two coordinates. So the load is on the $\mathbb{C}$ -field result; complementarity is consistency evidence, not the criterion.

The other two Role-1 nouns stay distinct. P07 (channels) is the *basis* the amplitude lives in — the distinguishable options, the pointer basis — separable from the amplitude values (channels are distinct even at equal bandwidth and phase). P02 (participation) is the primitive *relation* — a chain couples to a channel, here, now — the addressing the amplitude is content of, not the amplitude value. So the substrate nouns compress from four to three: **participation (a relation), channels (a basis), and the complex amplitude (their content)**.

5. Time-Homogeneity Belongs to the Process; Space and Time Do Not Share One Homogeneity

The arena appears to have two “no privileged spot” members — spatial homogeneity (P03: no privileged locus) and time-homogeneity (P13: no privileged moment). Two things need to be said, and they must be kept separate because they have very different strength.

The load-bearing, falsifiable claim: no unified spacetime homogeneity exists at the substrate level. The tempting move is to merge P03 and P13 into one spacetime-translation symmetry (the continuum’s Poincaré invariance read back as primitive). ED cannot: in ED time carries the arrow (commitment) and space does not — there is an arrow primitive (P11) and no spatial-arrow primitive. So “homogeneity of space” and “homogeneity of time” are homogeneities of two structurally different axes, not one symmetry in two guises, and a unified spacetime-translation symmetry is *emergent*, not primitive — ED’s substrate carries **less** symmetry than its smooth limit. This is the section’s real content, it is falsifiable (§10.4), and it rests entirely on one point of failure: **P11 being the sole asymmetric axis**. If that holds, the non-merge holds.

The reclassification (weaker, and honest about it): P13 is a property of the process, not an arena primitive. Here is the correction to an earlier, circular version of this argument. One *cannot* derive P13 from P11 by observing that “commitment is one fixed rule,” because calling the rule “one fixed rule” — the same at every moment — is already asserting time-homogeneity; that route is question-begging, and the position paper factors P13 out *as a separate primitive* precisely because P11’s collapse-content does not entail it. What can be said honestly is a reclassification, and it comes straight from the canonical text: Paper_087 states P13 as a predicate **of the dynamical primitive** — “*the substrate’s dynamical primitive is invariant under time translation.*” So P13 is, by its own definition, a property of the process-law (that the one law is time-translation invariant), **not** a fact about a separate temporal stage the way P03 is a fact about the spatial stage. It therefore belongs on the process side of the ledger, not in the arena — which removes it from the arena count ($5 \rightarrow 4$) as a *reclassification*, not a derivation. Once it is on the process side, the “merge with P03” is not even well-posed: a property of the process cannot merge with an arena primitive. So the trivial reading of the non-merge falls out for free; the non-trivial, falsifiable claim is the one above.

A scope note that survives either way: P13 is about the *laws*, not the *state*. The arrow may give time

a special *initial* moment (a first commitment), but a special initial state is a boundary condition, not a law-inhomogeneity — so the arrow’s possible beginning is a separate initial-condition matter and does not bear on P13, which is law-invariance.

6. The One That Does Not Reduce (Yet): Three Dimensions

Three-dimensionality (P06) is the hardest arena member, and the honest verdict is that it is **selected, not derived** — it stays a primitive. Three arguments land on 3, but they are not three derivations, and the paper should not present them as such. **Two of the three read 3 off observed physics rather than deriving it:** the internal amplitude dimension (a gauge family’s channels are mutually-orthogonal complex vectors, and the max in $\mathbb{C}^d$ is d , so the *observed* families $\{1,2,3\}$ fix $d = 3$ — a read-off of the gauge content), and the holographic reach law (only $d = 3$ gives the *observed* $g \sim 1/b$ — a read-off of gravity). Only the third, the linking argument, is a genuine derivation *candidate*. That the three coincide on 3 is real internal coherence, but two of them are anchored on measured content, so the honest status is selection, not derivation.

The linking argument is the one genuine *derivation candidate*, and it routes 3D back to commitment. A link (two loops threaded) is the only structure that holds an order against continuous rearrangement, and stable linking of one-dimensional loops exists in exactly three dimensions (2D cannot form a link; 4D unravels any link — rigorous topology). Read against the primitives, its premises are commitment-side: “committed order must be stable against continuous rearrangement” is commitment’s irreversibility (a committed order is permanent, hence topologically protected), and “1D loops” are the forward-causal chains. So the candidate is: $\{1D \text{ chains} + \text{committed order topologically stable} + \text{stable 1D-linking unique to 3D}\} \implies d = 3$ — which, *if it holds*, forces three-dimensionality from commitment needing to hold stable topological records of 1D chains.

The candidate rests on one open premise: that ED holds its committed order by linking *specifically*. Its status is half-settled. The cheap alternative — order held by node labels or timestamps — is **ruled out** on the certified simulator: relabel the substrate with physics fixed and the committed order barely moves (Kendall $\tau \approx 0.96$ versus 0.5 for a label-scramble; the committing-node set is invariant). What this measures is precisely that the order is **not label-held**; concluding it is *geometry-held* is an inference that holds only if labels and geometry exhaust the options, and the residual 4% movement is not zero — so the honest reading is the negative one (labels ruled out), not yet the positive one. And the structure that would hold it is reachable (the multi-chain participation graph can be intrinsically linked). What remains open is isolating that the holder is *linking* specifically, which requires a three-dimensional substrate to even ask, and so is not testable on the certified two-dimensional simulator — an honest hard limit, shared with the open curvature-emergence derivation. So P06 stays a primitive, with a commitment-connected, conditional route back.

7. The Floor and the Capstone

Collecting the audit, the one reduction, and the one reclassification: the substrate nouns compress $4 \rightarrow 3$ (§4); time-homogeneity moves from the arena to the process side (§5); the arena’s remaining members are spatial. The honest minimal **floor** is the eight items:

3 substrate (participation, channels, complex amplitude) + 4 arena (spatial homogeneity, 3D, scale, kinds) +
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That box is the floor. The two-word form below it is a **mnemonic for the floor, not a replacement of it**:

The shortest *naming* of Event Density is commitment and participation — the sole process, and the primitive relation it acts on. This names the two most fundamental *roles*; it is explicitly **not** a claim that channels, the amplitude, and the four-member spatial arena collapse into “participation.” Section 4 argued the opposite (channels and amplitude stay distinct from the participation relation), and that stands. “Participation” in the two-word form is standing in, as a mnemonic, for the whole noun-side that commitment addresses; the honest count remains the eight-item box.

With that scope fixed, two features are worth stating. First, even the mnemonic is a *pair*, not a single primitive — faithful to ED’s relational stance that a relation is prior to an isolated thing (participation is already a relation: a chain, to a channel, here, now; commitment is the act on that relation). Second, and the capstone:

ED has a spatial stage but no separate temporal stage. The remaining arena is purely spatial — three dimensions, spatial homogeneity, scale, kinds — with no arrow. Time contributes no *arena* primitive: its two aspects are commitment’s direction (P11) and the law-invariance of that one process (P13, a process-side property, §5). There is no “time” sitting in the arena beside space, because what we call time is commitment happening.

This is why the two-word naming is apt and not an oversimplification: one does not need a third word for a temporal stage, because there is no temporal stage — the arena is spatial, and time is the process. It remains a *naming* of roles; the floor is the box.

8. Discussion: Commitment as the Keystone of ED’s Physics

The reduction explains a pattern visible across ED’s derivations: commitment is not merely one primitive among many but the single load-bearing one, because it is the only process. Followed to their roots, ED’s physical structures are commitment acting:

- **Time and irreversibility** are commitment directly.
- **Measurement and the pointer basis** — commitment resolves only in the channel basis, so ED is a preferred-basis theory because the arrow selects the basis (einselection is primitive here).
- **The measurement-repeatability (covering-law) content** — irreversibility locks a committed channel, which is the repeatability a quantum-logic reconstruction needs.
- **Matter/antimatter (C)** — first-arrival commitment selects the charge conjugation, the one chirality-adjacent thing ED fixes natively.
- **Gravity’s causal direction** — the one-way black-hole horizon versus the two-way cosmological one is set by the arrow.
- **Information preservation** — the same irreversibility that forbids reversal forbids erasure: information is preserved as the permanent record of every commitment.

- **The single hypercharge $U(1)$** — polarity is phased against the one external flow, the commitment flow, so there is one gauged $U(1)$ because there is one arrow to phase against.

Each of these has its own support elsewhere in the program; assembled, they are the physical face of §3’s structural claim that commitment is the sole process. This synthesis is an **account**, not a theorem (preamble 4): it explains *why* commitment is everywhere without claiming the other primitives are derivable from it — they are its objects and its stage, and the paper has argued (§3–§6) exactly where commitment stops.

One guard against a rhetorical illusion the list can create: **each job is only as strong as its own paper, and several of the seven are themselves interpretation-tier** (that einselection is primitive, that matter/antimatter is native, that gravity’s direction and the single- $U(1)$ reference are the arrow’s). The list is a *pattern*, not a compounding of evidence — seven conditional accounts do not sum to one hard result. Read it as motivation for the pivot choice (§2), not as independent confirmation of it.

9. Limitations (honest)

- **The reduction is structural, not a derivation of the primitives.** The primitives remain postulated (Paper_087); this audits their internal dependencies. The count-reducing moves are of two different kinds and neither is a clean derivation: the §4 amplitude fusion is *inherited* (grounded on the C-field result of the reconstruction paper, and only as strong as that reconstruction-tier result), and the §5 time-homogeneity move is a *reclassification* (P13 belongs with the process by its canonical definition), explicitly **not** a derivation of P13 from commitment — an earlier “one fixed rule $\implies$ time-homogeneous laws” version was circular and is retracted here.
- **The audit is P11-pivoted** (§2). It cannot establish that commitment, rather than another primitive, is the natural pivot; that is an input. Its pivot-independent content is where commitment *fails* to absorb a primitive.
- **Three-dimensionality is not resolved.** P06 stays a primitive; its route back to commitment is a candidate with an open premise (§6), gated on the same curvature-emergence derivation that is open across ED’s geometric arc.
- **“Commitment is the sole process” is an audit result, not a proof of completeness.** It rests on reading each primitive against P11’s canonical statement; a different canonical statement of a primitive could shift a facet/independent call.
- **The keystone synthesis is an account.** §8 assembles per-job results into a unifying picture; it is not a single derivation and is labelled interpretation.
- **Values are untouched.** Nothing here reduces the empirical value layer; the minimality is of primitive structure only.

10. Falsification Criteria

10.1 A Role-1 noun is not presupposed by commitment

Falsifier sentence: *A demonstration that participation, channels, or the amplitude can be stated without commitment presupposing them — or that one of them is in fact a facet of commitment — would overturn the three-role split (§3).*

10.2 Bandwidth and phase are independent primitives

Falsifier sentence: *A demonstration that bandwidth and phase are jointly sharp (no complementarity) or that the amplitude field is not $\mathbb{C}$ would break the $4 \rightarrow 3$ fusion (§4) and restore them as two primitives.*

10.3 Time-homogeneity is an arena fact, not a process property

Falsifier sentence: *A demonstration that P13 (time-homogeneity) is a fact about a temporal stage independent of the process-law — the way P03 is a fact about the spatial stage — rather than a property predicated of the dynamical primitive, would falsify the §5 reclassification and return P13 to the arena.*

10.4 Space and time share one primitive homogeneity

Falsifier sentence: *A demonstration that ED's substrate carries a single unified spacetime-translation symmetry at the primitive level (not merely emergently) would falsify the space/time asymmetry claim (§5) and collapse the arrow's distinction between the two axes.*

10.5 Three-dimensionality is genuinely derived (or genuinely free)

Falsifier sentence: *Either a completed derivation of P06 (closing the linking premise) or a demonstration that no consistency argument selects 3 would move P06 off its current “selected, not derived” status (§6).*

11. Position Statement

The canonical thirteen primitives of Event Density organize into three roles — the things commitment acts on, the arena it runs on, and commitment with its facets — and, under a P11-pivoted audit, commitment (P11) is the sole process primitive. One reduction and one reclassification tighten the set: bandwidth and phase fuse into one complex amplitude (grounded on the inherited $\mathbb{C}$ -field result), and time-homogeneity is moved from the arena to the process side (P13 is, by its canonical definition, a property of the dynamical primitive — a reclassification, not a derivation of P13 from commitment). Time-homogeneity does not merge with spatial homogeneity — ED's time is arrowed and its space is not, so no unified spacetime symmetry exists at the substrate level; Poincaré invariance is emergent. The honest floor is 3 substrate + 4 arena + 1 commitment; its shortest *naming* is the pair **commitment and participation** (the process and the relation it acts on), a mnemonic for the floor, not a claim that the eight items collapse to two. ED has a spatial arena but no separate temporal one: time is the process.

What this paper claims. The three-role split and commitment-as-sole-process (structural, P11-pivot disclosed); the $4 \rightarrow 3$ amplitude fusion (grounded, inherited from the $\mathbb{C}$ -field result); the time-homogeneity reclassification (P13 is a process-side property, not an arena fact); the space/time asymmetry (no unified substrate homogeneity, resting on P11 being the sole asymmetric axis); the eight-item floor and the spatial-stage capstone.

What this paper does not claim. No derivation of the primitives from a deeper layer; no reduction to a single primitive (the two-word form is a mnemonic, not the floor); no *derivation* of time-homogeneity from commitment (an earlier circular version is retracted; the move is a

reclassification); no derivation of three-dimensionality (selected, with a conditional candidate); no reduction of empirical values; and the keystone synthesis is an account, not a theorem that does not compound.

Series context. The minimality companion to the position paper (Paper_087): where the position paper enumerates and postulates the thirteen, this paper audits their internal dependencies and reports the compact pair the reduction lands on. It consolidates the reduction program (the role audit, the amplitude fusion, the three-dimensionality test, and the time-homogeneity reclassification) into one cold-reader statement of what Event Density is, at bottom: the pair of *roles* — commitment and participation — naming an eight-item floor.

Appendix — Glossary and Reader Map

Glossary

- **Commitment (P11).** The irreversible collapse of a chain’s multi-channel participation to a single channel, with un-selected phases randomized. ED’s sole process primitive.
- **Participation (P02).** The primitive relation: a chain couples to a channel, at a locus, at a time. The sole thing commitment acts on.
- **Complex amplitude (P04 + P09).** $\sqrt{b} \cdot e^{i\pi}$ on a channel — bandwidth and phase as two real readings of one complex number, fused because the amplitude field is $\mathbb{C}$ (the inherited $\mathbb{C}$ -field result), with complementarity as corroboration.
- **Channels (P07).** The distinguishable options / pointer basis the amplitude lives in; a distinct primitive from the amplitude value.
- **Arena.** The stage commitment runs on: spatial homogeneity (P03), three dimensions (P06), scale (P08), kinds (P10). Purely spatial after §5.
- **Facet.** A primitive whose content is definitionally part of another; here chains (P01) and transport (P05) are facets of commitment, and P12’s descent-*direction* is a partial facet (its functional form is not). Time-homogeneity (P13) is not a facet but a *process-side property* — reclassified out of the arena (§5), not derived from commitment.
- **Selected, not derived.** A primitive picked out by consistency arguments (multiple independent ones landing on the same value) without being reduced to other primitives — the status of three-dimensionality.

Reader map

Intuition. Strip Event Density to its smallest honest description and two things remain: an act (commitment) and a relation it acts on (participation). Everything dynamical is the act; the amplitude and channels are what the relation carries; space is the stage. Time is not a third thing on the stage — time is the act happening.

Where the arrow stops. The value of this audit is as much in where commitment *fails* to reduce things as where it succeeds. It cannot absorb its own nouns (participation, channels, amplitude) or its spatial stage. It does absorb the entire temporal side of the arena. That asymmetry — space is a stage, time is the process — is the load-bearing, falsifiable content.

Where to look next. - *The primitives enumerated and postulated:* the position paper (Paper_087). - *The complementarity and $\mathbb{C}$ -field results the amplitude fusion rests on:* the quantum-

logic reconstruction. - *The open bridge that would settle three-dimensionality: the curvature-emergence arc.*

Open work (declared): close the three-dimensionality linking premise (gated on curvature emergence); test whether participation (P02) and channels (P07) could further merge into one relational primitive; and formalize “commitment is the sole process” beyond the primitive-statement audit.

End of Paper (Commitment and Participation).

Substrate-evaluation arc. A P11-pivoted minimality audit of the canonical thirteen primitives: they split into the things commitment acts on, the arena it runs on, and commitment with its facets; commitment is the sole process. Bandwidth and phase fuse into one complex amplitude (grounded on the inherited $\mathbb{C}$ -field result); time-homogeneity is reclassified from the arena to the process side (a property of the one law, not derived from commitment); space and time share no substrate homogeneity. Honest floor: 3 substrate + 4 arena + 1 commitment. Shortest naming (a mnemonic, not the floor): commitment and participation, over a purely spatial stage — ED has a spatial arena but no separate temporal one, because time is the process. Three-dimensionality stays selected, not derived, with a conditional route back to commitment.